

ResolveIQ Platform Redesign

Rev 3 touches only the surface: an operational Dashboard, a three-pane Investigation Workspace, enterprise-scale search, and a consistent visual language. Every architectural decision from rev 2 — Product Intelligence, Ingestion Pipeline, Recommendation Engine, Match Reasoning, Playbook Engine, the eight-module nav, the sprint sequence — is carried forward unchanged.

Status Draft — expected final review **Supersedes** RFC rev 2 **Architecture** Approved, unchanged — §02–§06
Changed this rev §07–§21 (UX only)

○○ What real data broke

Unchanged — carried forward for context.

INGESTION

Binary files parsed as UTF-8

.docx, .xlsx, and .jpg uploads were decoded as plain text, polluting entity extraction and the semantic index.

RANKING

Keyword overlap beat domain understanding

A Get Load Profile / firmware-DCW mismatch was out-ranked by an unrelated RF mesh timeout case on shared generic words.

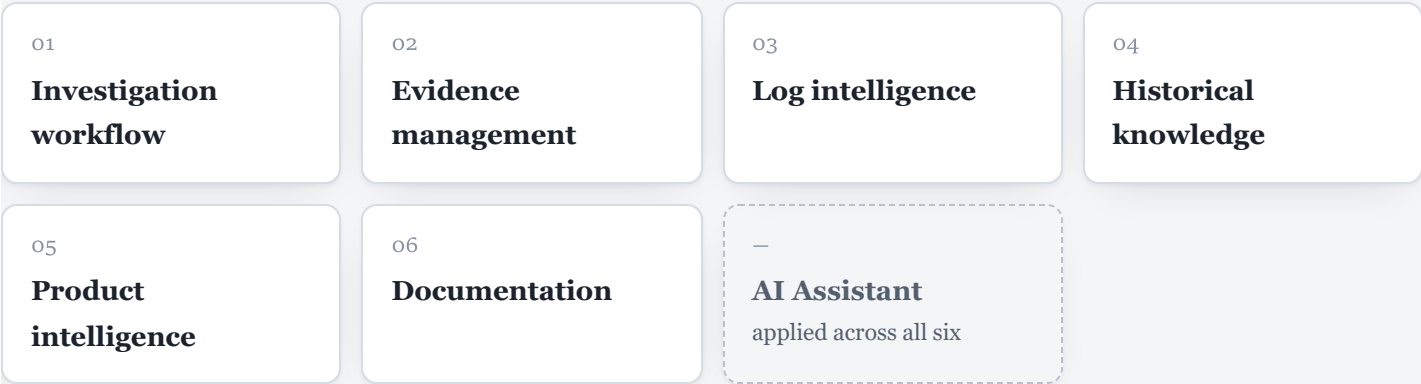
WORKFLOW

One scrolling page, no investigation shape

No navigation, tabs, or progress state — a single page holds a form submission, not an investigation.

01 Platform philosophy

Unchanged: **ResolveIQ is not an AI chatbot.** AI Assistant remains one module of eight; the six pillars below remain the product.



02 Architecture unchanged

Product Intelligence Engine sits as a cross-cutting knowledge layer beneath the six processing engines — it doesn't process evidence, it answers structured questions the Recommendation Engine asks of it. Every UX addition in this revision is new UI surfacing this same set of engines — see the scope map in §07 for the line-by-line mapping.

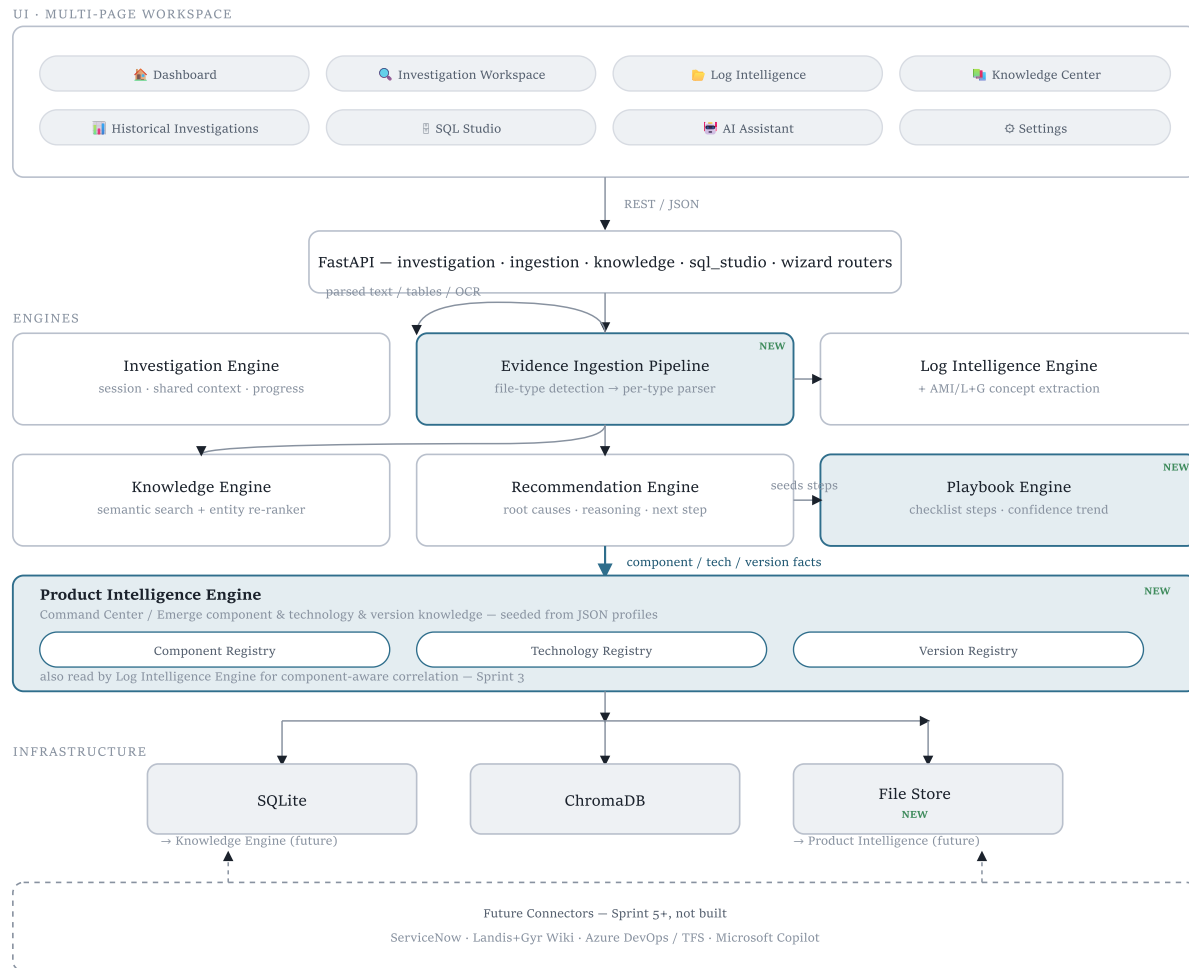


Fig. 1 — Product Intelligence Engine (accent) sits beneath the processing engines and feeds the Recommendation Engine directly. Approved in rev 2, unchanged.

Nothing in §08–§16 requires a new engine, a new datastore, or a framework change. Where a UX item needed *any* backend addition, it's a small field or enum on an existing domain model — called out inline, and totalled in §20.

03

Product Intelligence Engine

unchanged

Component Registry, Technology Registry, Version Registry — one engine, three registries, so lookup/link machinery isn't triplicated. Rev 3's Knowledge Center (§13) and Historical Investigations filters (§14) are new views over this same data, not new registries.

Component Registry — one profile per architecture component

CommandProcessorHost Command Center · service	
RESPONSIBILITIES Receives meter commands from the scheduler, validates against current DCW, dispatches to the mesh network	RELATED COMPONENTS CommandPayloadProcessorHost, MsgProcCmdRspHost, Scheduler
TYPICAL FAILURES Firmware/DCW mismatch after upgrade without GEI	RELATED LOGS CommandProcessorHost.log, CommandPayloadProcessorHost.log
KNOWN BUGS → known_bugs filtered by component	DOCUMENTATION → documentation filtered by component
INVESTIGATION PLAYBOOKS GLP/DCW mismatch triage, command timeout triage	COMMON SQL SELECT * FROM CommandLog WHERE CommandLogId = ?

Technology Registry & Version Registry

TECHNOLOGY	COMMON ISSUES	REQUIRED LOGS
RF Mesh	Mesh hop degradation, router firmware regressions	Collector command log, mesh health log
RF Mesh IP	IP routing table desync after topology change	Collector command log, routing table dump
Wi-SUN	Border router handoff failures	Wi-SUN border router log
Cellular	Carrier connectivity drops, SIM provisioning	Device Hub connection log
PLC	Line noise, phase coupling loss	Collector PLC signal log
Blended	Fallback-path misrouting between technologies	Collector command log (all technologies present)

PRODUCT · VERSION	KNOWN BEHAVIOR
Command Center 8.x	GEI required after every firmware push; not automatic
Command Center 9.x	GEI auto-triggered on firmware upgrade completion — a version mismatch invalidates an 8.x match

04

Evidence Ingestion Pipeline

unchanged

Nothing reaches entity extraction or the semantic index until it has been through the parser that understands its format.

FILE TYPE	PARSER	LIBRARY	OUTPUT
<code>.log .txt .csv .json .xml</code>	Text / structured parser	<code>stdlib</code> <code>csv</code> / <code>json</code> / <code>xml</code>	Plain text (+ table/tree)
<code>.docx</code>	Word document parser	<code>python-docx</code>	Paragraph & table text
<code>.xlsx</code>	Spreadsheet parser	<code>openpyxl</code>	Per-sheet cells + structured rows
<code>.pdf</code>	PDF parser	<code>PyMuPDF</code>	Page text, OCR fallback
<code>.jpg .png</code>	Image OCR parser	<code>pytesseract</code> + Pillow	OCR'd text, flagged low-confidence
<code>.evtx</code>	Windows Event Log parser	<code>python-evtx</code>	One structured event per record
<code>.zip</code>	Archive parser	<code>stdlib</code> <code>zipfile</code>	Recurses per entry → child Evidence
anything else	Unsupported file handler	—	Binary blob only, never decoded as UTF-8

Still flagging: OCR needs the tesseract binary on the host — best-effort, not a hard dependency, for Sprint 2.

05

Recommendation + transparency

unchanged

Hybrid semantic + entity-overlap ranking, plus a rule-based Match Reasoning factor list. Rev 3 gives this a permanent home in the Workspace's right pane (§09) instead of a single tab.

SEMANTIC ONLY	HYBRID RE-RANK
1 Meter comm. timeout, COLL-14 score 0.79 generic AMI vocabulary overlap	1 GLP / firmware-DCW mismatch score 0.91 exact command_type, host_component match

✓ Same component — CommandProcessorHost
✓ Same firmware — 13.63
✗ Different product version — CC 9.2 vs 8.4

06 Navigation flow unchanged

Dashboard → Wizard → Workspace, with global sidebar access to all eight modules. The Workspace's internal layout changes (\$09); which module leads to which does not.

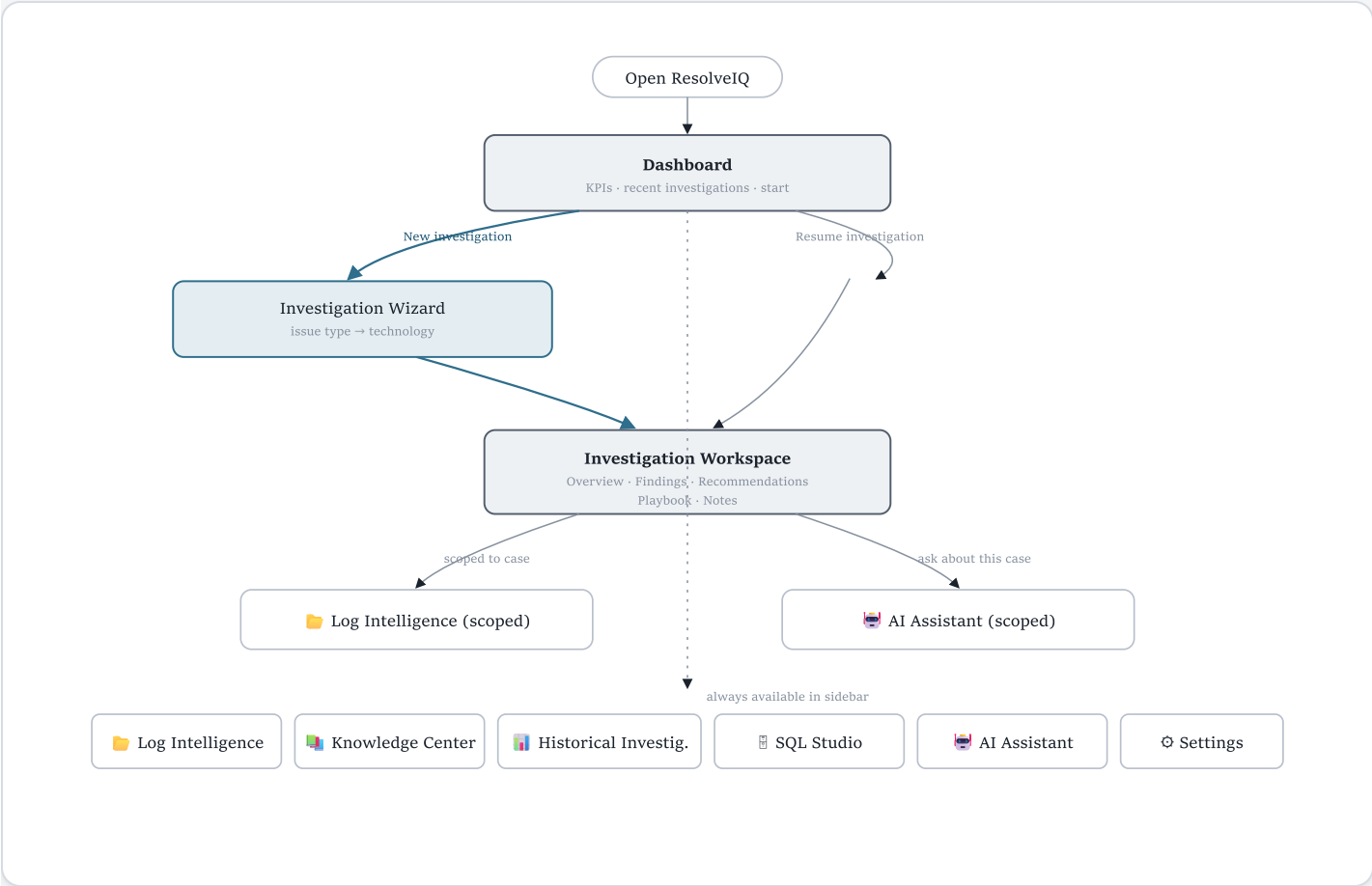


Fig. 2 — Accent path is the primary flow for engineers who start with only a symptom, not an ID. Approved in rev 2, unchanged.

07 UX pass — scope map

Every item from your feedback, mapped to where it lands and whether it needs any backend change.

#	YOUR REQUIREMENT	SECTION	BACKEND IMPACT
1	Dashboard as operational hub	§08	None — reads existing Investigation/Knowledge Engine lists
2	Three-pane investigation console	§09	None — layout only
3	Knowledge Center as central knowledge base	§13	Documentation gets a <code>category</code> tag; rest is existing collections
4	Enterprise-scale historical search filters	§14	4 new fields on <code>HistoricalInvestigationRecord</code>
5	Full SQL Studio structure	§15	New <code>SavedQuery</code> domain model (still no execution)
6	Investigation Timeline	§10	New <code>TimelineEntry</code> , appended by existing Investigation Engine calls
7	Documentation Generators	§11	None — 2 more template types on the existing AI Assistant templating
8	Investigation status lifecycle	§12	<code>InvestigationStatus</code> enum: 4 → 6 values
9	UI polish / visual language	§16	None — CSS injection into Streamlit
10	Persistent Investigation Summary card	§09	2 new fields (<code>assigned_engineer</code> , <code>resolved_at</code>) on <code>InvestigationSession</code>

o8 Dashboard — the operational hub

Azure Portal / Jira shape: everything an engineer needs to resume work, above the fold, no click required.

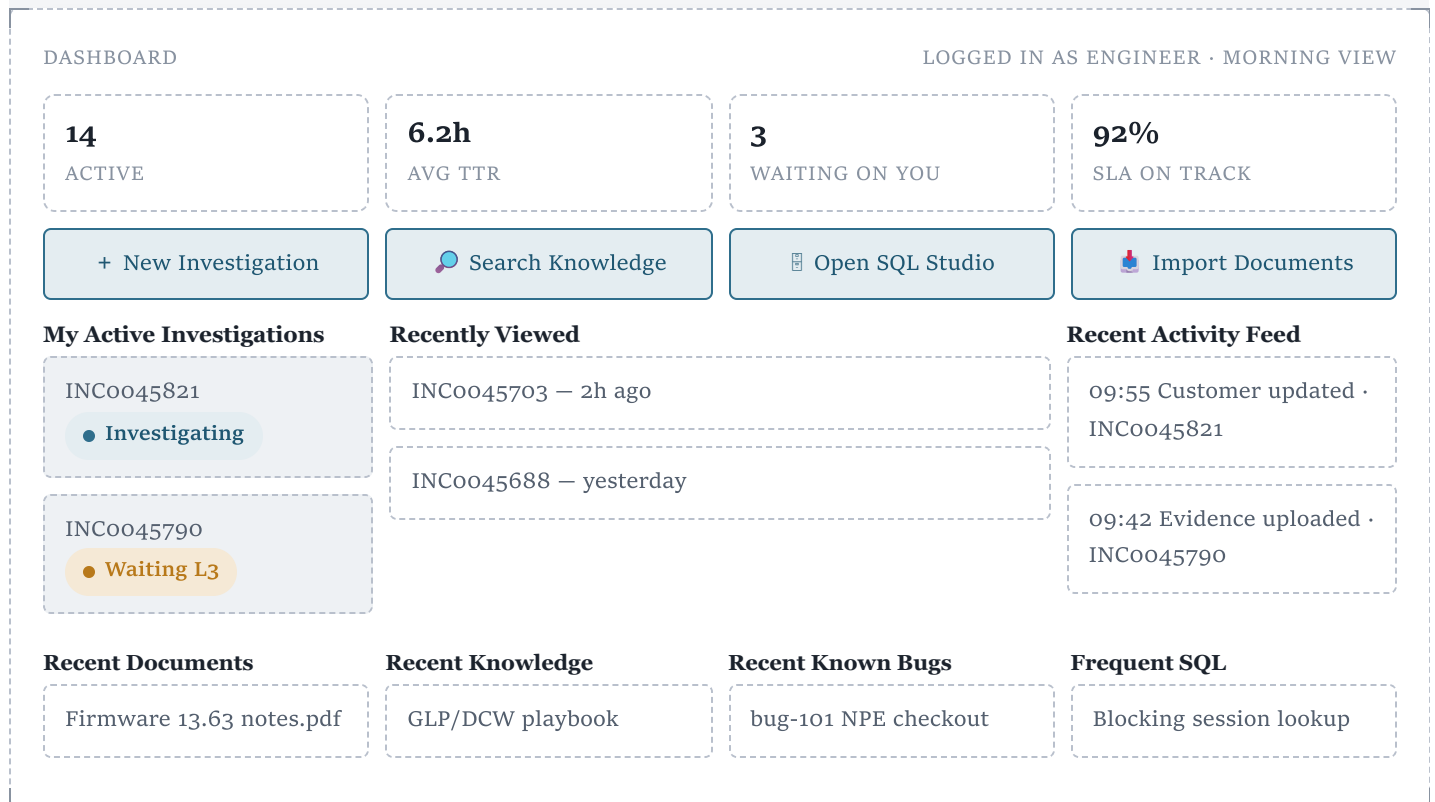


Fig. 3 — All 8 of your requested panels present; KPI row and Quick Actions stay fixed at top so nothing requires scrolling to reach.

09 Investigation Workspace — three-pane console

VS Code shape: a persistent Summary Card, a left explorer for evidence, a center workbench for reasoning, and a right assist rail that never requires leaving the center pane to consult.

Persistent Investigation Summary — your item 10

CUSTOMER Acme Utilities	ISSUE GLP → Ping response	PRODUCT · VERSION Command Center 9.2	TECHNOLOGY RF Mesh	STATUS ● Investigating
ENGINEER J. Alvarez	CONFIDENCE 91%	TIME SPENT 1h 12m	HYPOTHESIS Firmware/DCW mismatch	

Sourced entirely from existing fields — InvestigationSession, merged_entities, and Recommendation.overall_confidence — plus two new fields noted in §07.

Three panes

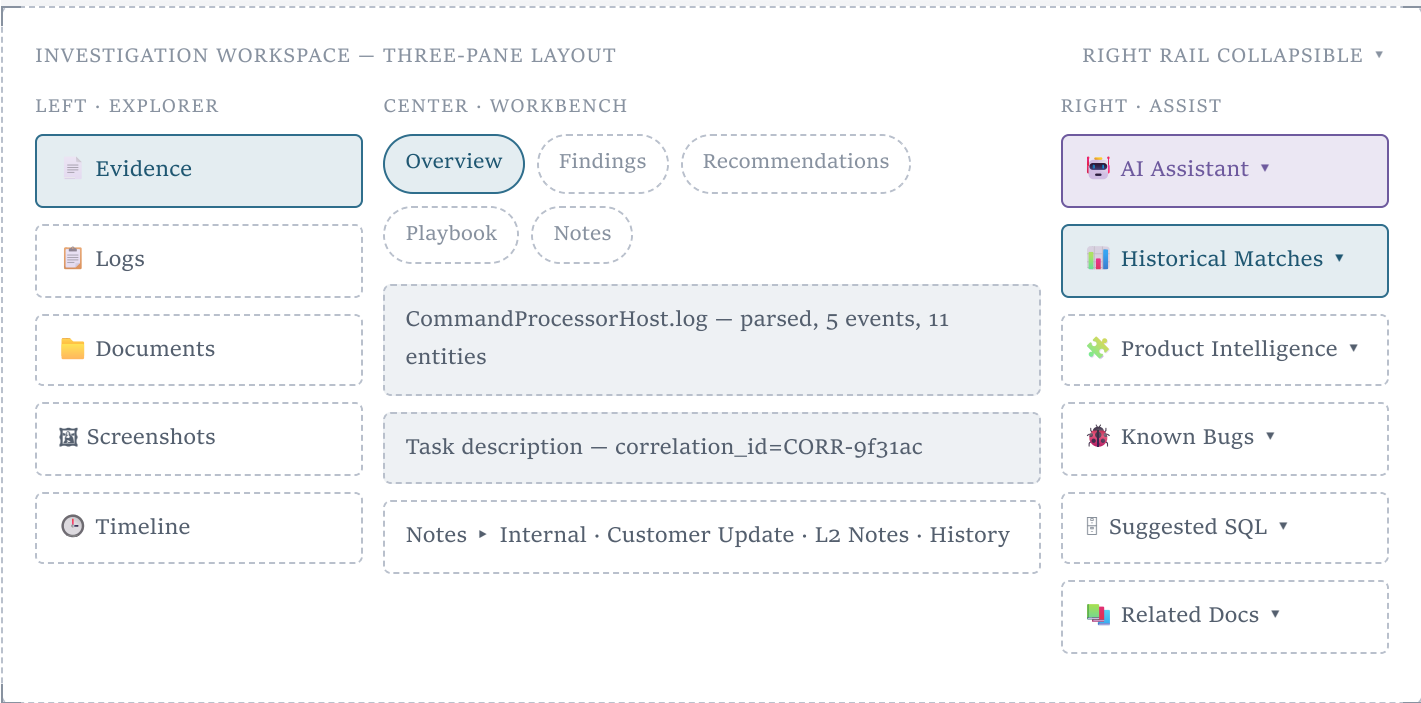


Fig. 4 — Left explorer swaps center content by evidence type; right rail sections collapse (▾) independently so the engineer keeps only what's relevant open.

Recommendations, in the right rail

The Match Reasoning panel from rev 2 (§05) now lives permanently in the right rail's "Historical Matches" section rather than a tab that has to be opened:

✓ **Same component** — CommandProcessorHost

✓ **Same firmware** — 13.63

✗ **Different product version** — CC 9.2 vs 8.4

Progress, above the Playbook tab

Investigation progress



Collect Logs



Analyze



Match



Known Bug

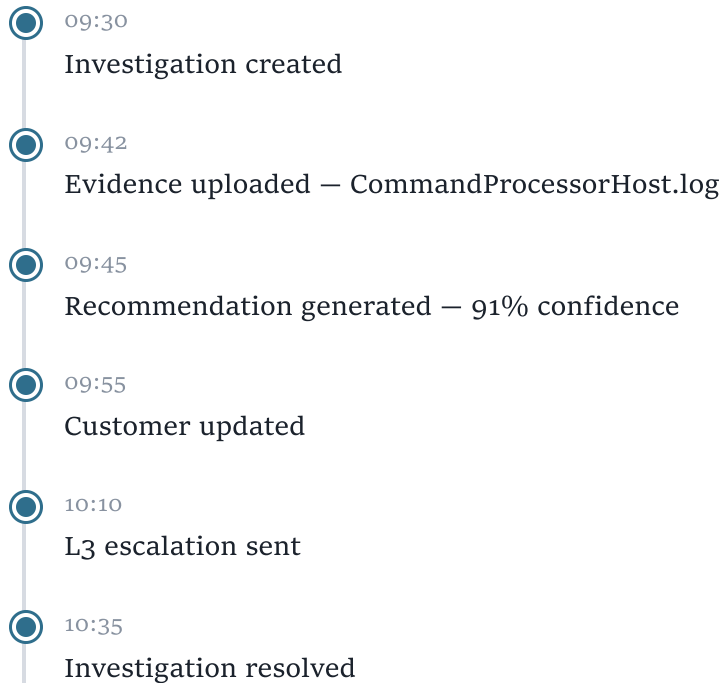


Resolution

Unchanged mechanism from rev 2 — now rendered inside the Playbook tab instead of a separate section.

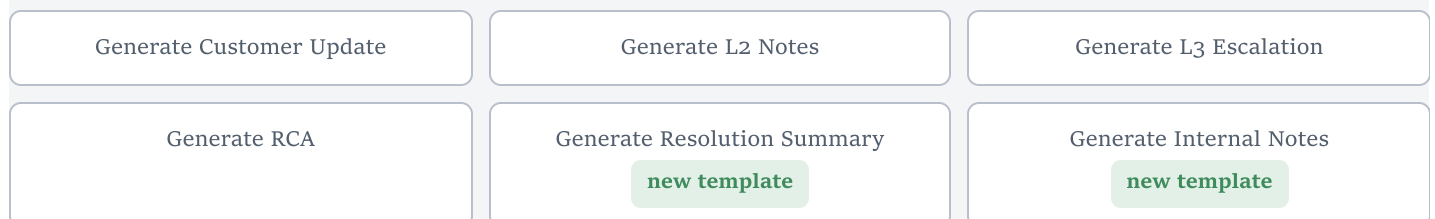
10 Investigation Timeline — your item 6

Auto-built, not manually maintained. Every state-changing call the Investigation Engine already makes (start_investigation, add_evidence, a future update_status) appends one TimelineEntry — no new engine, just a row alongside Evidence in the existing repository.



11 Documentation Generators — your item 7

Not a new engine — the AI Assistant module's templated-draft capability (already scoped, no-LLM, in rev 2) surfaced as first-class buttons inside the Workspace's Notes tab, plus two more template types.



Each button fills a template from the Summary Card + Findings + top Match Reasoning — the same data already on screen, not a new data source.

12 Investigation Status — your item 8

InvestigationStatus grows from 4 values (open/in_progress/resolved/closed) to 6, matching real triage stages. This is the one enum change in this revision.

STATUS	BADGE	MEANING
Open	● Open	Created, not yet started
Investigating	● Investigating	Actively worked
Waiting for Customer	● Waiting	Blocked on customer response
Waiting for L3	● Waiting	Escalated, blocked on L3
Resolved	● Resolved	Fix identified/applied, pending close
Closed	● Closed	Done

Feeds the Summary Card's status cell (§09) and the Dashboard's "by status" breakdown (§08) directly.

13 Knowledge Center — your item 3

Almost everything you asked for already exists in rev 2's architecture — this section is mostly a new browsing UI over data the Knowledge Engine and Product Intelligence Engine already store, organized by category instead of one flat search box.

CATEGORY	SOURCE
Documentation, Runbooks, Playbooks, FAQs, Release Notes	Knowledge Engine's <code>documentation</code> collection, grouped by a new <code>category</code> tag
Product Components, Firmware / DCW	Product Intelligence Engine — Component & Version Registries (\$03)
Technologies	Product Intelligence Engine — Technology Registry (\$03)
Known Bugs	Knowledge Engine's <code>known_bugs</code> collection — unchanged
L3 Investigations	Historical investigations filtered by severity — §14's filters, reused

KNOWLEDGE CENTER

Documentation

Product Components

Technologies

Firmware / DCW

Known Bugs

Release Notes

Runbooks

Playbooks

L3 Investigations

FAQs

 search across all categories

Playbook: GLP/DCW mismatch triage

Runbook: Collector command timeout

Release notes — firmware 13.63

Fig. 5 — Left rail is the 10 categories you specified; center is the same semantic search from rev 2, scoped to the selected category.

14 **Historical Investigations — your item 4**

Product, Version, Technology, and Component filters map directly onto Product Intelligence Engine data already extracted per investigation. Customer, Issue Type, Severity, Status, Tags, and Date Range are 4 new fields added to HistoricalInvestigationRecord (Tags and a domain/category field already existed in rev 1).

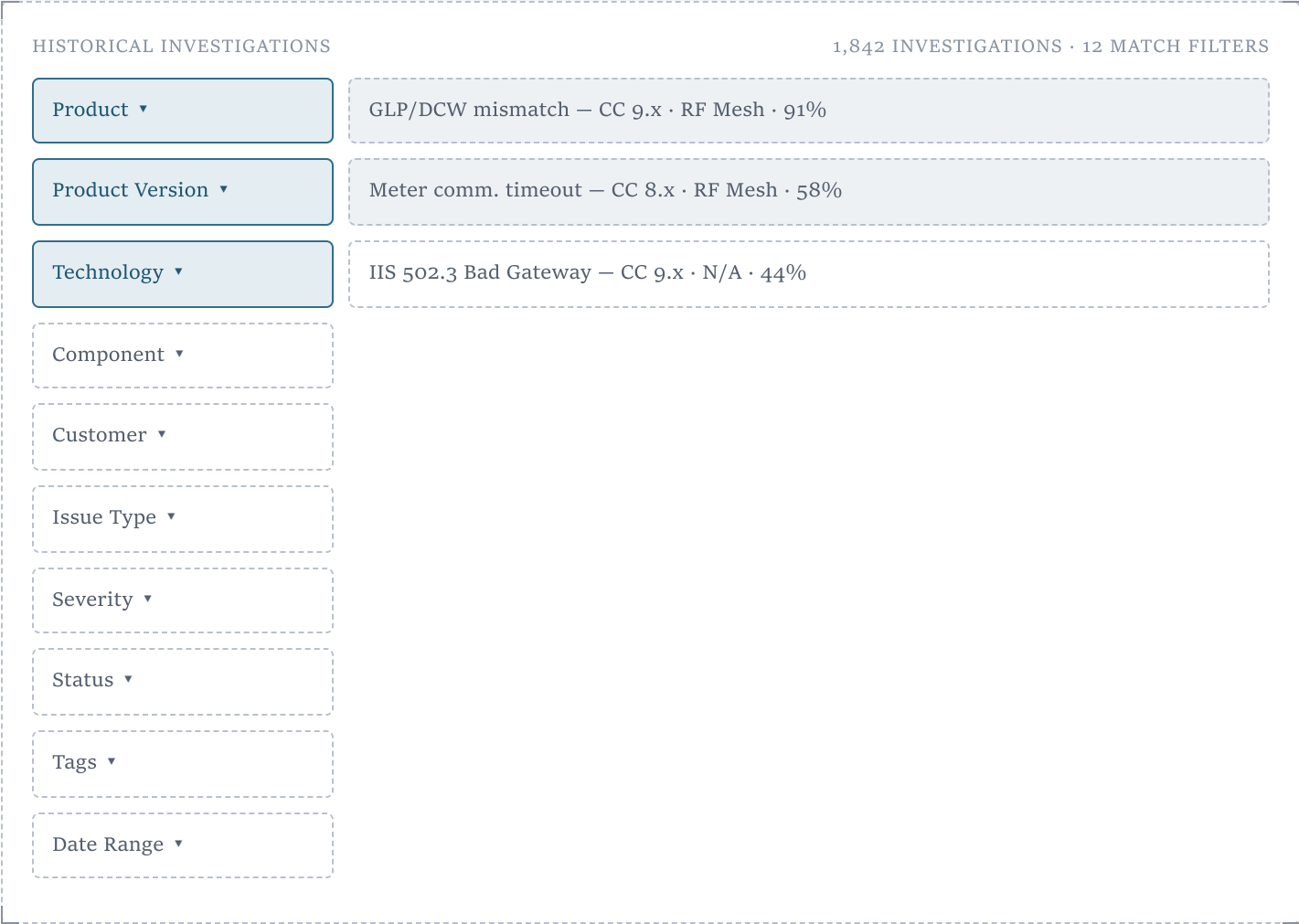


Fig. 6 — Filter panel narrows before search runs, not after — the "1,842 investigations" framing is the enterprise-scale case you flagged.

15 SQL Studio — your item 5

Suggested SQL and Query Generator are the existing suggested_sql mechanism from the Recommendation Engine (rev 1), surfaced here too. Saved/Recent/Favourite Queries and the Query Library are one new SavedQuery domain model. Query Explanation is a pre-written description per template, not a live analyzer. Execution stays out of scope, as you specified.

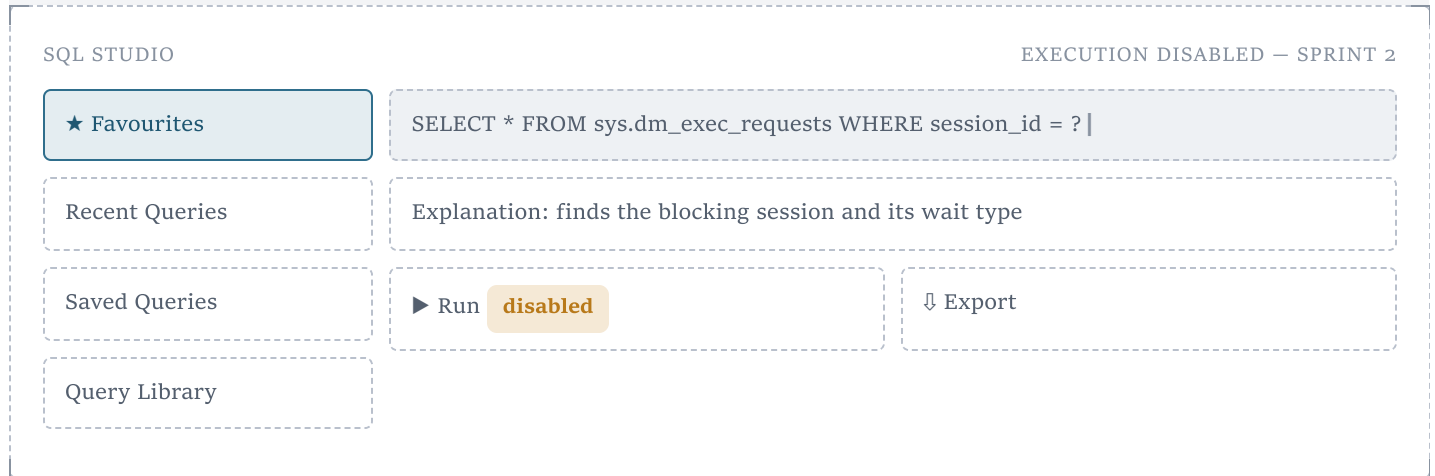


Fig. 7 — Full structure ships in Sprint 2; only the Run button stays disabled pending a future sprint's DB-execution decision.

16 Visual language — your item 9

Streamlit stays the framework (per rev 2's approved architecture) — this section is what "Azure Portal / Linear" polish means concretely inside it, so the ambition is honest about what's CSS and what would need more.

ASK	HOW IT'S DONE IN STREAMLIT	EFFORT
Cards, spacing, typography, colors	One injected CSS stylesheet (<code>st.markdown</code>) — tokens matching this doc's palette	Low
Status badges	Small HTML component, colors keyed to §12's status table	Low
Three-pane / multi-column layout	<code>st.columns</code> — native, exactly this shape	Low
Collapsible right-rail sections	<code>st.expander</code> — native	Low
Modern icons	Consistent icon set (not mixed emoji) via inline SVG component	Medium
Sticky Summary Card / sidebar	CSS <code>position:sticky</code> on the container — works, some Streamlit re-render quirks to test	Medium
Splunk-grade log viewer (rev 2, §Log Intelligence)	Custom HTML/JS component — already scoped in rev 2	Medium-High
True drag-resizable panes	Not native to Streamlit; would need a custom component or a frontend rewrite	Out of scope this sprint

Net: the "professional platform" feel is achievable as CSS + native Streamlit layout primitives for everything except drag-resize, which nobody asked for explicitly and isn't blocking the "not a standard Streamlit app" goal.

17 Revised folder structure

Additions from rev 2 are unchanged; below are rev 3's small domain additions and the new UI files.

```
app/domain/
├─ ... (rev 2 unchanged: ingestion, playbook, wizard, product_intelligence)
├─ enums.py          (modified) InvestigationStatus: 4 → 6 values
├─ investigation.py  (modified) + assigned_engineer, resolved_at
├─ evidence.py       (modified) HistoricalInvestigationRecord + customer, issue_type, severity, status
├─ timeline.py       # NEW — TimelineEntry
├─ sql_studio.py     # NEW — SavedQuery, QueryTemplate

app/engines/investigation/engine.py (modified) appends TimelineEntry on state changes

data/sample_knowledge/
├─ historical_investigations.json (modified) + customer/issue_type/severity/status per record

ui/
├─ Home.py + pages/*.py (unchanged — 8 pages)
├─ theme.css            # NEW — injected stylesheet: tokens, cards, badges, spacing
├─ components/
│   └─ ... (rev 2 unchanged)
│   └─ summary_card.py  # NEW — persistent header, §09
│   └─ status_badge.py  # NEW — §12
│   └─ timeline_view.py # NEW — §10
│   └─ doc_generator_panel.py # NEW — §11
```

18 Module responsibilities

MODULE	SPRINT 2 SCOPE	CHANGED THIS REV
 Dashboard	Full — operational hub, §08	Expanded from KPI strip to 8-panel hub
 Investigation Workspace	Full — three-pane console, §09	Tabs-only → three panes + Summary Card
 Log Intelligence	Search/filter/entities/errors — correlation/threads Sprint 3	Unchanged from rev 2
 Knowledge Center	10 categories over existing collections, §13	Flat docs → categorized repository
 Historical Investigations	Full search + 10 filters, §14	Search box → filter panel
 SQL Studio	Full structure, execution disabled, §15	Templates only → full structure
 AI Assistant	Scoped chat + 6 generator templates, §11	4 templates → 6
 Settings	Read-only	Unchanged

19

Roadmap

unchanged

Sprint sequence from rev 2 stands. This revision adds UI scope inside Sprint 2's existing boundary — no sprint moves.

SPRINT	THEME	EXPLICITLY EXCLUDED
2	Redesign foundation + this UX pass	Live connectors, thread reconstruction, cross-file correlation, SQL execution
3	Log Intelligence depth + SQL Studio execution	Live ServiceNow
4	Knowledge Center bulk import + AI Assistant	Live Wiki/ADO sync, external LLM calls
5	Connectors	Copilot integration
6	Production hardening	—

20 **Sprint 2 plan — additions this rev**

Rev 2's file list stands in full. Below is what's added for the UX pass — small domain fields plus UI.

DOMAIN ADDITIONS
app/domain/enums.py modified — InvestigationStatus: open, investigating, waiting_customer, waiting_l3, resolved, closed
app/domain/investigation.py modified — + assigned_engineer, resolved_at
app/domain/evidence.py modified — HistoricalInvestigationRecord + customer, issue_type, severity, status
new app/domain/timeline.py TimelineEntry
new app/domain/sql_studio.py SavedQuery, QueryTemplate (+explanation text)
app/engines/investigation/engine.py modified — append TimelineEntry on every state-changing call
UI
new ui/theme.css Injected stylesheet — cards, badges, spacing, type scale from this doc's tokens
new ui/components/summary_card.py, status_badge.py, timeline_view.py, doc_generator_panel.py §09–§12 components
ui/pages/1_Dashboard.py modified — 8-panel hub layout, §08

ui/pages/2_Investigation_Workspace.py

modified — three-pane via st.columns, §09

ui/pages/4_Knowledge_Center.py

modified — 10-category rail, §13

ui/pages/5_Historical_Investigations.py

modified — 10-filter panel, §14

ui/pages/6_SQL_Studio.py

modified — full structure, execution disabled, §15

TESTS

new tests/test_timeline.py, test_status_transitions.py, test_sql_studio_models.py

Same coverage discipline as prior sprints

21 Sign-off

New this revision

- Confirm the three-pane Workspace layout (§09) reads correctly — left explorer swaps center content, right rail is independent/collapsible
- Confirm the visual-language approach (§16): CSS injection + native Streamlit layout, no frontend rewrite this sprint

Carried over — still open

- OCR as best-effort, not a hard dependency — §04
- Custom Streamlit component for the log viewer — §16
- No LLM/external API calls in AI Assistant for Sprint 2, including the 2 new Documentation Generator templates — §11

